HU ZHENTAO

AI / ML Research Engineer - LLM Applications, Speech Interaction, and Wireless Sensing

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PROFILE

AI and signal-processing researcher with project experience across LLM speech interaction, audio algorithm evaluation, sensor-based deep learning, and wireless human activity recognition.

Built datasets, reproduced vision-model baselines, designed experiments, tested model fine-tuning routes, and authored first-author publications including an IJCNN 2025 paper.

Product-aware technical profile for AI product roles that need a PM who can reason about model behavior, evaluation, deployment tradeoffs, and applied user value.

CORE SKILLS

- Machine learning: Python, PyTorch, Transformer/CNN families, ResNet, Swin Transformer, ViT variants, benchmark reproduction, experiment design.
- LLM and speech: Whisper, Llama3-8B, Qwen, DeepSeek, ASR/TTS chains, audio-data fine-tuning tests, PEFT vs. full-parameter fine-tuning.
- Optimization and alignment exposure: SFT, DPO, PPO, GRPO concepts and comparative experimentation in sensor/LLM reasoning contexts.
- Signal and tools: time-series signal processing, mmWave radar sensing, MATLAB, C/C++, Adobe Audition, HFSS, LaTeX, Office, Figma.

RESEARCH AND ENGINEERING EXPERIENCE

Wireless Human Activity Recognition with mmWave Signals - UCAS / ICT
Jun 2023 - Dec 2025

- Collected and organized raw mmWave radar data, defined the experimental plan, and completed thesis-related experiments for human activity recognition.
- Built a dataset covering 6 daily activity classes and 20 participants; the sample scale exceeded the public datasets used as project references at the time.
- Reproduced and compared mainstream visual-model baselines, including ResNet, Swin Transformer, ViT, and related variants, to build a multi-dimensional benchmark.
- Explored preprocessing and feature-extraction methods inspired by wavelet and fractional Fourier transforms to improve end-to-end activity recognition robustness.
- Extended the research direction toward LLM reasoning over sensor data, including Qwen2.5-based inference, SFT on a smaller Qwen model, and comparisons across DPO/PPO/GRPO style reinforcement-learning routes.

Voice Interaction Model Based on Llama3-8B
Oct 2024 - Dec 2024

- Built a pre-research speech interaction chain with whisper_large_v2 for ASR, Llama3-8B for dialogue reasoning, and tacotron2-DDC-GST for TTS.
- Tested Qwen and DeepSeek model families for audio-data fine-tuning behavior, comparing parameter-efficient fine-tuning and full-parameter fine-tuning.

Huawei Consumer BG - Audio Algorithm Engineer Intern
Jul 2024 - Oct 2024

- Studied spectrograms, 3A algorithms, and audio algorithm evaluation methods; built a model prototype and collected data based on ICASSP 2021 nn-ref related work.
- Participated in endpoint AEC testing on iPhone and Huawei phones, developing practical experience in audio algorithm evaluation and issue localization.

APPLIED AI PRODUCT EXPOSURE

- At an early-stage startup in the emotional companion AI direction, worked on AI product delivery across multi-agent business advisory assistants, companion digital humans, AIGC video generation, AI comic workflows, and AI training products.
- Translated technical capabilities into product scenarios, acceptance criteria, deployment materials, budget/procurement documents, and vendor coordination workflows.

EDUCATION

- University of Chinese Academy of Sciences, M.S. in Electronic Information, Institute of Computing Technology, Sep 2022 - Jan 2026.
- Xidian University, B.S. in Electronic Information, Sep 2018 - Jun 2022.

PUBLICATIONS

- Time-domain multi-scale analysis based wireless human activity recognition method, first author.
- T-FrFT: Fractional Fourier Transform based Discriminative Feature Extraction Method for Human Activity Classification, IJCNN 2025, first author.